Objective Type Questions

7. If $x^2-y^2 = 16$ and $(x+y) = 8$ then $(x-y)$ is _____

(A) 8 (B) 3 (C) 2 (D) 1

8. $\frac{(a+b)(a^3-b^3)}{(a^2-b^2)}$ = _____

(A) a^2-ab+b^2 (B) a^2+ab+b^2 (C) $a^2+2ab+b^2$ (D) $a^2-2ab+b^2$

9. $(p+q)(p^2-pq+q^2)$ is equal to _____

(A) p^3+q^3 (B) $(p+q)^3$ (C) p^3-q^3 (D) $(p-q)^3$

10. $(a-b)=3$ and $ab=5$ then a^3-b^3 = _____

(A) 15 (B) 18 (C) 62 (D) 72

11. $a^3+b^3 = (a+b)^3$ – _____

(A) $3a(a+b)$ (B) $3ab(a-b)$ (C) $-3ab(a+b)$ (D) $3ab(a+b)$

3.7 Factorisation

Expressing any number as the product of two or more numbers is called as **factorisation.** The number 12 can be expressed as the product of prime factors like $12 = 2 \times 2 \times 3$. This is called prime factorisation. How will you factorise an algebraic expression? Yes. Expressing an algebraic expression as the product of two or more expressions is called the Factorisation.

Note

A number which is divisible by 1 and itself (or) A number which has only 2 factors are called prime numbers. Example: 2, 3, 5, 7, 11, ...

A number which has more than 2 factors are called composite numbers.Example: 4, 6, 8, 9, 10, 12, ...

For example, (i) $a^2-b^2 = (a+b)(a-b)$ Here, $(a+b)$ and $(a-b)$ are the two factors of a^2-b^2

(ii) $5y+30 = 5(y+6)$, Here 5 and $(y+6)$ are the factors of $5y+30$

Any expression can be Factorised as $(1) \times$ (expression)

For example, a^2-b^2 can also be factorised as

$$(1)\times\left(a^2-b^2\right) \text{ or } (-1)\times\left(b^2-a^2\right)$$

because '1' is a factor for all numbers and expressions

So, when we factorise the expressions, follow the suitable type of factorisation given below to get two or more factors other than 1. Stop doing the factorisation process once you have taken out all the common factors from the expression and then list out the factors.

Type 1: Factorisation by taking out the common factor from each term.

Example 3.18

Factorise: $4x^2y+8xy$

Solution:

We have, $4x^2y+8xy$ This can be written as,

$= (2\times2\times x\times x\times y) + (2\times2\times2\times x\times y)$

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